

one can compute the TF for the query terms and return a ranked list of documents in which the query terms appear most frequently. If this suffices to return the most relevant documents, why do we need the measure of Inverse Document Frequency (IDF)? Can you illustrate with an example why IDF is needed in information retrieval? If you ignore IDF for relevant document retrieval, will it impact the precision or recall or both?

- (7) You are given two arrays and are asked to find the common elements between them. (a) Assume that the two arrays only contain integer elements. (b) Assume that the two arrays contain floating point values. Will your solution for (a) work for (b) as well?
- (8) Different activation functions can be employed for the neurons such as rectilinear, tan-h, sigmoid, etc. Can you explain the factors involved in the choice of activation functions? Are there any criteria that a function needs to satisfy for it to be considered as an activation function in a neural network trained using back propagation?
- (9) You are given an array A of N integers. You are also given a number X. You are asked to find the element in A that is closest to X. You are given the constraint that you should not sort the array A. (a) Can you write a C program to solve this problem? (b) Now you are told that the set S consists of floating point values instead of integers; how will you modify your solution for (a) to solve (b)?
- (10) You are given a set of documents and you are asked to perform clustering of these based on document level similarity. You are familiar with different clustering algorithms such as 'kmeans', so you are confident that you can do the clustering. However, the interviewer now adds a second part to the problem where he wants you to label the clusters. How will you generate labelled clusters? What are the means of generating meaningful labels for the clusters? Can the clusters be labelled post the clustering process or prior to the clustering process or during the cluster process itself? Describe your choice of the solution and why you made that particular choice?
- (11) Spatial applications are getting developed in large numbers these days. Consider a taxi-fleet management company where trajectories of its taxis need to be stored in a database. Spatial applications should be able to query the database for a specific trajectory or segments of a trajectory. Given a set of location points, it should be possible to retrieve all the taxis whose trajectories had overlaps over the location points. What kind of database will you use for storing trajectory data? For speeding up query performance, what kind of indexing will you perform for trajectory data?
- (12) You are given the problem of disambiguating entities in a given set of documents. For example, for the word 'Apple', you will need to disambiguate between the use of the word 'Apple' as a corporate entity versus as a fruit. A more complicated example would be the following:

S1: The White House announces its intention to support the newly formed government in Monaco.

S2: The New Year Party will be held at White House, and will be hosted by Melania Trump.

In sentence S1, White House is an entity that represents the American government. In sentence S2, White House is a location. Can you come up with an algorithm that can disambiguate such entity mentions to their appropriate types based on the context?

(13) In solving classification and regression problems in machine learning, great emphasis is laid on feature engineering, where the right set of features are carefully selected among all the features of the given data samples. For instance, consider a linear regression problem where each sample of input data has four features X_1, X_2, X_3 and X_4 . X_1 and X_2 are linearly dependent on X_3 . X_1 is linearly independent of all the other features. For example, assume that the regression problem is to predict house prices. X_1 is the type of locality, X_2 is the size of the house, X_3 and X_4 are the size of the bedroom and size of the external yard. You have created two classifiers G1 and G2, such that G1 uses all four features and G2 uses only X_1 and X_2 . Will G2 be more accurate than G1? What are the problems with G1 in comparison with G2?

(14) Given a classifier, can you define the terms sensitivity and specificity of your system? Let us assume that you have built a classifier that can predict whether a tumor is benign or malignant based on certain tumor characteristics. Your intention is to build a classifier that can detect all true malignant tumors, even if some of the benign tumors get falsely reported as malignant. Which of the two metrics — sensitivity or specificity — should you try to improve in this case? Now assume that you have been asked to use a classifier to predict whether a student will pass the examination or not. Your system should be designed such that it does not report a student who has failed as having passed, even if it mis-predicts certain passing students as potential failure students. In that case, which of the two metrics (sensitivity or specificity) should you try to improve in this case?

(15) Given a non-empty array of integers, return the 'k' most frequent elements. Give multiple solutions that optimise for space and time complexity. How will your solution change if your input is a continuous data stream?

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